

Efficient Algorithm for Expanding Theoretical Electron Densities in Canterakis–Zernike Functions

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An algorithm for the efficient computation of Canterakis–Zernike moments of theoretically computed molecular electron densities and rotationally invariant Fingerprint indices derived from them is reported. The algorithm is suitable for any density expressed in terms of Gaussian- or Slater-type functions within the Linear Combination of Atomic Orbitals framework at any level of computation. Electron density is expressed as a one-center expansion of real regular spherical harmonics times radial factors by means of translation techniques, which

facilitates the efficient computation of the moments in terms of a single one-dimension numerical integration. The performance of the algorithm is analyzed showing that the computation of radial factors in the quadrature points is responsible for almost all computational time. The procedure is applicable to any density obtained with standard packages for molecular structure calculations. © 2018 Wiley Periodicals, Inc.

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Introduction

Molecular similarity measures are more than just a way of comparing two molecules. In essence, they are a way to establish the concept of neighborhood in the molecular space.^[1,2] In general, it is difficult to, given a single molecule, establish which other molecules are in its vicinity in the set of all possible molecules. Ultimately this problem has its origins in the fact that there are no coordinate systems for the space of molecules. We do not have a set of numbers which takes a specific combination of values for each and every molecule such that no particular combination of values repeat,^[2–4] much like how the Cartesian coordinate system maps three values (*x*, *y*, *z*) to each point in the Euclidean space and no two points have the same particular set of values for the Cartesian triplet.

Once a metric is obtained, it is possible to define, in the chemical space, other structures such as surfaces, smooth functions, paths, trajectories or local curvatures. All of these features could help chemists better navigate the chemical space looking for patterns, or even allowing the entirety of calculus to be applied to a space of molecules. It is no wonder, then, that there is an ongoing effort to establish a reliable and general measure of similarity.^[5–10]

One of the strategies to measure the similarity between two molecules is to somehow transform each molecule in a vector of traits, called Fingerprints, and then compare these vectors with regular linear algebra operations, an approach that has been widely used in QSAR studies and molecular screening strategies.^[11–16] Applying such a strategy, naturally, only changes the problem from comparing molecules to turning molecules to vectors. One possible approach to making a Fingerprint vector is to create a binary vector and have each bit in that vector represent the presence or absence of certain chemical groups or properties in the molecule, for example, whether it has rings, or if it has anything other than carbon and hydrogen, or whether it has a carbonyl moiety.

It is clear that such fingerprints are not meant to be based on a chemical coordinate system but rather are very domain specific because the decision on what features are going into the Fingerprints is arbitrary. As such, there is no way to guarantee that there will not be two molecules with the same vector, except for very narrow sets of molecules. A more general approach is to construct Fingerprints that can be used throughout a very wide set of molecules. This is where the Fingerprint concept meets the rotationally invariant indices of the Canterakis–Zernike expansion.

The Zernike functions are a set of orthonormal functions in the unit disk that have been devised by Fritz Zernike in 1934 to describe diffraction properties in phase contrast microscopes.^[17] Written in polar coordinates the odd Zernike polynomials have the form $R_m^n(r)\sin(m\varphi)$ and the even ones are written as $R_m^n(r)\cos(m\varphi)$ in which the radial part $R_m^n(r)$, also called the Zernike radial polynomial is given by:

$$R_m^n(r) = \sum_{k=0}^{\frac{n-m}{2}} (-1)^k \binom{n-k}{k} \left(\frac{n-2k}{2} - k \right) r^{n-2k} \quad (1)$$

for an even $n-m$ and 0 for an odd $n-m$. Since the Zernike polynomials form a complete set in the unit disk, any 2D image that can be mapped to the disk can be decomposed as a linear

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combination of Zernike polynomials whose coefficients are referred to as Zernike moments. Such a linear combination can be used to reconstruct the original image in an approximate manner, retaining the general features and losing the fine detail except in the limit of an infinite expansion. In addition, in his 1934 paper, Zernike showed how it was possible to use the moments of the expansion to construct indices that were invariant with the rotation in the plane of the disk, which allows for the comparison of images in any orientation.

In his technical report of 1999, Canterakis^[18] elaborated on the rotational invariance property of the Zernike polynomials as well as its resistance to noise and highlighted its potential importance to applications in computational vision. With this in mind, Canterakis presents a way to construct orthonormal functions in the unit sphere which are 3D analogs of the Zernike functions, such that any 3D scalar field can be decomposed in the Canterakis–Zernike basis much in the same way as 2D images are decomposable in terms of the disk polynomials. And in much the same way as the Zernike moments, the Canterakis–Zernike moments can be used to construct indices which are invariant under rotations in 3D space, allowing for scalar fields to be compared in any orientation.

The electron density is a scalar field in 3D, and thus appropriate for the decomposition in terms of the Canterakis–Zernike basis. Because the electron density is an observable of any molecule, the Canterakis–Zernike Fingerprint vector is not domain specific and can be obtained for a wide range of molecules by means of the same algorithm.

The use of such rotational invariants is not an entirely novel concept, expansion in terms of spherical harmonics were already being used widely in QSAR community^[19] to study the isosurfaces of various scalar fields relevant to the chemical properties of molecules such as electrostatic potential energy, solvent accessibility, the electron density or its Laplacian. Despite having the same property of generating rotationally invariant indices, the spherical harmonics expansions had some issues which were resolved with the Canterakis–Zernike expansion. After the publication of Novotni and Klein's algorithm^[20] for shape decomposition in computer vision, the usage of that technique to study molecular systems, particularly proteins, increased substantially.

The Canterakis–Zernike Fingerprints were shown to expand the qualities that made the spherical harmonics so reliable for that kind of study as well as solving some of its difficulties^[21] and already that same year it was used to study the surfaces of proteins by Sael et al.^[22,23] A year later, in 2009, Gunasekaran et al. showed that the Canterakis–Zernike fingerprints could be used to find reliable ligands to a protein in a database,^[24] later on the same research group used the expansion and its fingerprints to study changes between conformational states of proteins.^[25] In the same year Venkatraman et al. published a work in which the Canterakis–Zernike Fingerprints were used to compare ligands in an actual QSAR study.^[26]

One of the most relevant works came in 2011, when Yin and Dokholyan expanded on the work done by Sael et al. in 2008^[22] and instead of using the Fingerprints of a protein's surface to match it against a database of other proteins, they used the entire electron density, obtained experimentally by cryo-

electromicroscopy, against the then complete set of single chain entries in the Protein Databank.^[27] In their work, Yin and Dokholyan proposed the rotationally invariant indices as a way to find structures in the PDB that could match an electron density and then be used as template to resolve experimentally obtained electron density maps. Their work not only showed the capability of the algorithms to engage in these searches but it also set the standard for the use of these Fingerprints in the comparison of proteins. A year after that, Hosny and Hafez dedicated a work to the implementation of new algorithmical techniques,^[28] particularly because so far the present algorithm required the expensive computation of spatial grids on which to evaluate the density and the solution of expensive and error-prone three-dimensional integrals.

Although Hosny and Hafez have achieved great success in their approach, obtaining very successful speedups, their algorithm is still a general algorithm for any unstructured scalar field defined over a grid, and thus scales unfavorably with the number of points used, which determines the overall accuracy of the moments.

This work proposes an algorithm that takes into consideration the fact that most of the electron densities obtained from regular quantum chemical software packages are not unstructured fields, but rather linear combinations of atomic orbitals with known coefficients and analytical form, in the hopes of getting rid of the grid computation as a means to obtain more accurate moments. The procedure followed in the derivation of the algorithm is described in the "Methodology" section. In particular, the general expansion of molecular density in a ball is presented first (see section "General expansion of molecular density in a ball"), followed by the algorithm for translation of STO and GTO charge distributions (see section "Expansion of STO and GTO densities with translation techniques"), and its application for expansion in the particular case of Canterakis–Zernike functions (see section "Expansion of density in Canterakis–Zernike functions"). The performance of the algorithm and its dependence with the size of the system and the length of the expansions is analyzed in the "Results and Discussion" section, where the accuracy attained in the electron multipole moments of a big system is used to show the correctness and stability of the algorithm. This section finalizes with an example of the electron density reconstruction in terms of its Canterakis–Zernike expansion.

Methodology

Most Standard packages for molecular structure calculations, at any computational level, yield electron densities expressed as a linear combination of basis functions of either Gaussian- or Slater-type functions. The algorithm reported here is valid for both types of functions and can be discussed in general terms. Nevertheless, specific details for the translation of charge distributions in case of Gaussians will be given in text, whereas suitable references will be provided for those pertaining to Slaters.

General expansion of molecular density in a ball

In the Linear Combination of Atomic Orbitals (LCAO) framework, at any level of computation, the electron density, ρ , can be expressed as:

$$\rho(\mathbf{r}) = \sum_{r=1}^m \sum_{s=1}^m \rho_{rs} \chi_r(\mathbf{r}) \chi_s(\mathbf{r}) \quad (2)$$

in which χ are basis functions, and ρ_{rs} , the elements of the first-order density matrix, the summations running over the set of m basis functions.

In this work, we are interested in densities expressed in terms of normalized real Gaussian (GTO) or Slater-type functions (STO). In both cases, functions are defined as products of a radial part, depending only in the polar coordinate r , times an angular part, which can be expressed in terms of powers of Cartesian coordinates x , y and z , or spherical harmonics. As Cartesian coordinates are used only in GTO with low values of quantum numbers (usually up to d functions, i.e. $L = 2$), whereas spherical harmonics are used both for GTO and STO, we will restrict ourselves to deal here with angular functions consisting of spherical harmonics. In fact, this is not a serious limitation, since powers of Cartesian coordinates can be readily expressed as linear combinations of regular spherical harmonics.

With the above considerations, the so-called Gaussian *primitive* functions, depending on two quantum numbers (L , M) associated with the angular part, are defined as:

$$g_{LM}(\mathbf{r}) = \mathcal{N}_{\xi}^{r(\text{GTO})} r^L e^{-\xi r^2} \mathcal{N}_{LM}^{\Omega} z_L^M(\theta, \varphi) \quad (3)$$

and the commonly used Gaussian *contracted* functions are linear combinations of *primitives*:

$$\chi_{LM}^{(\text{GTO})}(\mathbf{r}) = \sum_{i=1}^{N_G} c_i g_{LM,i}(\mathbf{r}) = \left[\sum_{i=1}^{N_G} c_i \mathcal{N}_{\xi_i}^{r(\text{GTO})} r^L e^{-\xi_i r^2} \right] \mathcal{N}_{LM}^{\Omega} z_L^M(\theta, \varphi) \quad (4)$$

where N_G is the length of the contraction, index i runs over the primitive functions in the contraction, and the expansion coefficients, c_i , are chosen so that the radial part of the contraction remains normalized.

On the other side, Slater functions depend on the (L , M) quantum numbers as before plus a third quantum number, N , associated with the radial part, and are given by:

$$\chi_{NLM}^{(\text{STO})}(\mathbf{r}) = \mathcal{N}_{N\zeta}^{r(\text{STO})} r^{N-1} e^{-\zeta r} \mathcal{N}_{LM}^{\Omega} z_L^M(\theta, \varphi) \quad (5)$$

In both types of functions, $z_L^M(\theta, \varphi)$ are unnormalized real spherical harmonics:

$$z_L^M(\theta, \varphi) = (-1)^M P_L^{|M|}(\cos\theta) \begin{cases} \cos M\varphi & (M \geq 0) \\ \sin |M|\varphi & (M < 0) \end{cases} \quad (6)$$

and the radial and angular normalization factors in eqs. (3)–(5) are given by:

$$\mathcal{N}_{\xi}^{r(\text{GTO})} = 2^{L+1} \sqrt{\frac{\xi^L}{(2L+1)!}} \left[\frac{(2\xi)^3}{\pi} \right]^{1/4} \quad (7)$$

$$\mathcal{N}_{N\zeta}^{r(\text{STO})} = \left[\frac{(2\zeta)^{2N+1}}{(2N)!} \right]^{1/2} \quad (8)$$

and

$$\mathcal{N}_{LM}^{\Omega} = \left[\frac{(2L+1)}{2(1+\delta_{M0})} \frac{(L-|M|)!}{\pi (L+|M|)!} \right]^{1/2} \quad (9)$$

In the following, symbol χ will be used to denote GTO or STO basis functions indistinctly, unless otherwise explicitly stated.

Our goal in this work is to expand the electron density inside a ball of radius r^* in terms of functions centered at the coordinates origin, which are orthogonal in the ball. In particular, we will apply the procedure to Canterakis–Zernike functions and, for this purpose, we will choose as starting point a set of functions, $\mathcal{F}_{kl}^m$, which are orthonormal in the ball of radius unity:

$$\mathcal{F}_{kl}^m(\mathbf{t}) = \mathcal{R}_{kl}(t) \mathcal{N}_{lm}^{\Omega} z_l^m(\theta, \varphi) \quad (10)$$

where $\mathbf{t} \equiv (t, \theta, \varphi)$ stands for the spherical coordinates, and:

$$\langle \mathcal{F}_{kl}^m | \mathcal{F}_{k'l'}^{m'} \rangle_1 \equiv \int_0^1 dt t^2 \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\varphi \mathcal{F}_{kl}^m(\mathbf{t}) \mathcal{F}_{k'l'}^{m'}(\mathbf{t}) = \delta_{kk'} \delta_{ll'} \delta_{mm'} \quad (11)$$

The subindex 1 in the bracket means that integration is carried out in the unit ball.

To keep orthonormality of functions $\mathcal{F}_{kl}^m$ inside the ball of radius r^* , we redefine them as:

$${}_k \tilde{\mathcal{F}}_l^m(\mathbf{r}) \equiv \frac{1}{\sqrt{r^{*3}}} \mathcal{F}_{kl}^m(\mathbf{r}/r^*) \quad (12)$$

To verify that functions ${}_k \tilde{\mathcal{F}}_l^m$ thus defined are orthonormal in the ball of radius r^* :

$$\langle {}_k \tilde{\mathcal{F}}_l^m | {}_{k'} \tilde{\mathcal{F}}_{l'}^{m'} \rangle_{r^*} \equiv \int_0^{r^*} dr r^2 \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\varphi {}_k \tilde{\mathcal{F}}_l^m(\mathbf{r}) {}_{k'} \tilde{\mathcal{F}}_{l'}^{m'}(\mathbf{r}) = \delta_{kk'} \delta_{ll'} \delta_{mm'} \quad (13)$$

it is sufficient to replace ${}_k \tilde{\mathcal{F}}_l^m$ in terms of $\mathcal{F}_{kl}^m$ —eq. (12)—and make the change of variable $t = r/r^*$, thus eq. (13) is reduced to eq. (11). Now, the subindex r^* in the bracket means that integration is carried out in the ball of radius r^* .

The expansion of the electron density inside the ball of radius r^* in terms of functions ${}_k \tilde{\mathcal{F}}_l^m$ reads:

$$\rho(\mathbf{r}) = \sum_{k=0}^{\infty} \sum_{l=0}^{\infty} \sum_{m=-l}^l {}_k \omega_l^m {}_k \tilde{\mathcal{F}}_l^m(\mathbf{r}) \quad (14)$$

To obtain the expansion coefficients ${}_k \omega_l^m$ we multiply eq. (14) on the left by ${}_{k'} \tilde{\mathcal{F}}_{l'}^{m'}(\mathbf{r})$ and carry out the integration in the ball of radius r^* taking into account the orthonormality relation (13):

$$\begin{aligned} \langle k' \tilde{\mathcal{F}}_{l'}^{m'} | \rho \rangle_{r^*} &= \sum_{k=0}^{\infty} \sum_{l=0}^{\infty} \sum_{m=-l}^l k \omega_l^m \langle k' \tilde{\mathcal{F}}_{l'}^{m'} | k \tilde{\mathcal{F}}_l^m \rangle_{r^*} \\ &= \sum_{k=0}^{\infty} \sum_{l=0}^{\infty} \sum_{m=-l}^l k \omega_l^m \delta_{kk'} \delta_{ll'} \delta_{mm'} = k' \omega_{l'}^{m'} \end{aligned} \quad (15)$$

Replacing now eq. (2) in (15) and suppressing the primes for short, the expansion coefficients result:

$$\begin{aligned} k \omega_l^m &= \langle k \tilde{\mathcal{F}}_l^m | \rho \rangle_{r^*} \\ &= \sum_{r=1}^m \sum_{s=1}^m \rho_{rs} \langle k \tilde{\mathcal{F}}_l^m | \chi_r \chi_s \rangle_{r^*} \end{aligned} \quad (16)$$

Therefore, it is necessary to solve the integrals appearing in:

$$\langle k \tilde{\mathcal{F}}_l^m | \rho \rangle_{r^*} = \sum_A \sum_B \sum_{r=1}^{m_A} \sum_{s=1}^{m_B} \rho_{rs}^{AB} \int_0^{r^*} dr r^2 \int_0^\pi d\theta \sin\theta \int_0^{2\pi} d\varphi_k \tilde{\mathcal{F}}_l^m(\mathbf{r}) \chi_r^A(\mathbf{r}_A) \chi_s^B(\mathbf{r}_B) \quad (17)$$

where we have introduced the indices *A* and *B*, which label the centers of the basis set functions, to remark that integrals involve functions centered at different points: expansion functions which are centered at the coordinates origin, and basis functions which can be placed at centers different from origin, thus yielding two- and three-center integrals, which are expensive to compute.

Coefficients $k \omega_l^m$ can be taken as the basis to define fingerprints for molecular electron density. In particular, the rotationally invariant quantities:

$$F_{kl} = \sqrt{\sum_{m=-l}^l (k \omega_l^m)^2} \quad (18)$$

have been proposed for this purpose in case of Canterakis-Zernike expansions,^[27] and can be generalized to expansions in other type of functions.

Expansion of STO and GTO densities with translation techniques

To avoid the cumbersome and expensive calculation of integrals in (17), an alternative procedure based on the translation of products of pairs of basis functions previously reported^[29] can be profited. This procedure allows us to expand the product of two basis functions in a third point, here taken as the axes origin, as a series of radial factors times angular-normalized regular spherical harmonics, $\mathcal{N}_{lm}^{\Omega} r^l z_l^m(\theta, \varphi)$, centered at that point:

$$\chi^A(\mathbf{r}_A) \chi^B(\mathbf{r}_B) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \mathcal{N}_{lm}^{\Omega} r^l z_l^m(\theta, \varphi) f_{lm}^{(N_A)L_A M_A(N_B)L_B M_B}(r) \quad (19)$$

where the N_A and N_B quantum numbers only appear in case of STO basis sets.

The radial factors are easier to compute in a frame with a suitable orientation, with center *A* lying on the *z*-axis and center *B* on the *xz* plane, as shown in Figure 1. Taking this frame, an algorithm based on recurrence relations, which allows us to efficiently compute those factors for given values of *r*, has been reported.^[29] Usually, the set of values of *r* will be chosen to carry out a final one-dimension numerical integration on this coordinate in the interval $[0, r^*]$, as it will be commented below.

Although this procedure has been previously reported for STO,^[29] it can be used with GTO too. In this case, the radial factors can be directly obtained by combining the Gaussian product theorem, and the formula for translation of GTO to another center. This procedure was proposed by Kaufmann and Baumeister^[30] and yields:

$$\begin{aligned} e^{-\xi_A |\mathbf{r}-\mathbf{R}_A|^2} e^{-\xi_B |\mathbf{r}-\mathbf{R}_B|^2} &= \sum_{l=0}^{\infty} \sum_{m=-l}^l \mathcal{N}_{lm}^{\Omega} r^l z_l^m(\theta, \varphi) 2\pi e^{-\xi_A \xi_B R_{AB}^2 / (\xi_A + \xi_B)} \\ &\times \mathcal{N}_{lm}^{\Omega} z_l^m(\mathbf{P}) e^{-(\xi_A + \xi_B)(r^2 + P^2)} \sqrt{\frac{\pi}{(\xi_A + \xi_B) r P}} I_{l+1/2}[2(\xi_A + \xi_B) r P] \end{aligned} \quad (20)$$

where $R_{AB} = |\mathbf{R}_B - \mathbf{R}_A|$, $\mathbf{P} = (\xi_A \mathbf{R}_A + \xi_B \mathbf{R}_B) / (\xi_A + \xi_B)$, $P = |\mathbf{P}|$ and $I_{l+1/2}(z)$ are the corresponding Bessel functions of half-integer order—see^[31] eq. 8.467.

The fact that the orientation of axes in Figure 1 usually does not coincide with the molecular frame is not a real problem, since the rotation of regular real spherical harmonics can be performed also very efficiently.^[32] This rotation transforms each $r^l z_l^m(\theta, \varphi)$ into a linear combination of regular harmonics with the same *l* and several (usually all) compatible *m* values, what implies that radial factors in the rotated frame are just linear

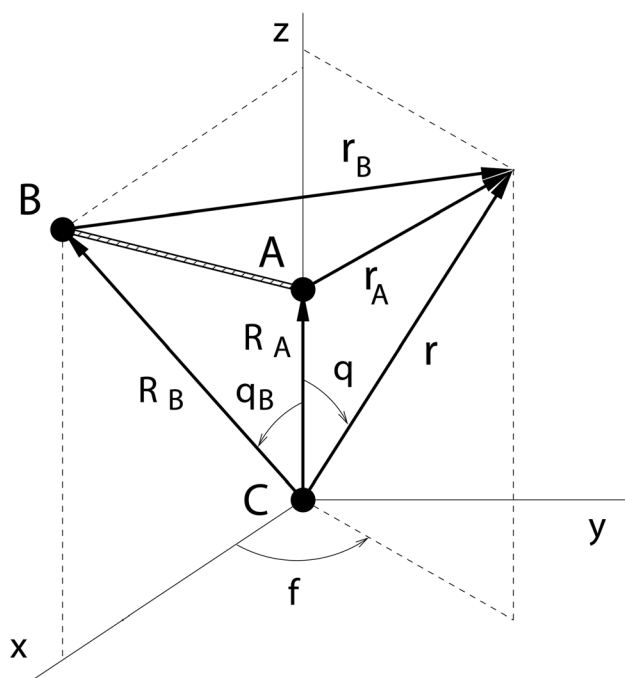


Figure 1. Frame for translation of two-center charge densities, $\chi^A(\mathbf{r}_A) \chi^B(\mathbf{r}_B)$, to a third center *C*.

combinations of the corresponding unrotated factors. Thus, introducing the rotation matrix, $\mathbf{D}^l$, which transforms the normalized real regular harmonics with a given l between both frames, eq. (19) can be rewritten as:

$$\begin{aligned}\chi^A(\mathbf{r}_A)\chi^B(\mathbf{r}_B) &= \sum_{l=0}^{\infty} \sum_{m=-l}^l [\mathcal{N}_{lm}^{\Omega} r^l z_l^m(\theta, \varphi)] f_{lm}^{(N_A)L_A M_A(N_B)L_B M_B}(r) \\ &= \sum_{l=0}^{\infty} (\mathbf{Z}^l)^t \mathbf{f}^l \\ &= \sum_{l=0}^{\infty} (\hat{\mathbf{Z}}^l)^t \mathbf{D}^l (\mathbf{D}^l)^t \hat{\mathbf{f}}^l\end{aligned}\quad (21)$$

where $\mathbf{Z}^l$ and $\mathbf{f}^l$ are column vectors containing, respectively, the angular-normalized regular spherical harmonics, $\mathbf{Z}_m^l = \mathcal{N}_{lm}^{\Omega} r^l z_l^m(\theta, \varphi)$, and the radial factors, $\mathbf{f}_m^l = f_{lm}^{(N_A)L_A M_A(N_B)L_B M_B}(r)$, in the molecular frame, whereas $\hat{\mathbf{Z}}^l$ and $\hat{\mathbf{f}}^l$ are the corresponding vectors in the frame oriented as in Figure 1. Therefore, we obtain the radial factors associated with the pair of basis functions in the molecular frame, $f_{lm}(r)$, from those computed in the oriented frame, $\hat{f}_{lm}(r)$, as:

$$f_{lm}^{(N_A)L_A M_A(N_B)L_B M_B}(r) = \sum_{m'=-l}^l \mathcal{D}_{m'm}^l \hat{f}_{lm'}^{(N_A)L_A M_A(N_B)L_B M_B}(r) \quad (22)$$

Proceeding in this way, for each pair of centers A and B , the radial factors of every pair of basis functions belonging to these centers are computed in the oriented frame, multiplied by the density matrix (rotated to that frame too), and accumulated. Once the radial factors have been added for the whole set of functions in centers A and B , the resultant radial factors are rotated back to the molecular frame and further accumulated to those coming from other pairs of centers. At the end of the process, the electron density can be expressed as a one-center expansion in the molecular frame whose terms are products of spherical harmonics times radial factors, $f_{lm}(r)$ (one per l, m), namely:

$$\rho(\mathbf{r}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \mathcal{N}_{lm}^{\Omega} r^l z_l^m(\theta, \varphi) f_{lm}(r) \quad (23)$$

and the expansion coefficients of the full density inside the ball of radius r^* , can be computed by:

$$\begin{aligned}{}_k\omega_l^m &= \langle {}_k\tilde{\mathcal{F}}_l^m | \rho \rangle_{r^*} \\ &= \sum_{l'=0}^{\infty} \sum_{m'=-l'}^{l'} \int_0^{r^*} dr r^2 \int_0^{\pi} d\theta \sin\theta \int_0^{2\pi} d\varphi {}_k\tilde{\mathcal{F}}_l^m(\mathbf{r}) [\mathcal{N}_{l'm'}^{\Omega} r^{l'} z_{l'm'}^m(\theta, \varphi)] f_{l'm'}(r)\end{aligned}\quad (24)$$

The integration of the angular part is trivial, because of the orthonormality of the spherical harmonics, and the coefficients ${}_k\omega_l^m$ can be computed by a one-dimension integral:

$${}_k\omega_l^m = \int_0^{r^*} dr r^2 \tilde{\mathcal{R}}_{kl}(r) r^l f_{lm}(r) \quad (25)$$

where $\tilde{\mathcal{R}}_{kl}(r)$ is the normalized radial part of function ${}_k\tilde{\mathcal{F}}_l^m(\mathbf{r})$, related to the radial factor of function $\mathcal{F}_{kl}^m(\mathbf{r}/r^*)$ by:

$$\tilde{\mathcal{R}}_{kl}(r) = \frac{1}{\sqrt{r^{*3}}} \mathcal{R}_{kl}(r/r^*) \quad (26)$$

The integral over the radial coordinate can be solved numerically, using the tabulated values of the radial factors at the selected values of r , as mentioned before. Thus the expansion coefficients in eq. (14) result:

$${}_k\omega_l^m = \sum_{i=1}^{\alpha} w_i^* \tilde{\mathcal{R}}_{kl}(r_i) r_i^{l+2} f_{lm}(r_i) \quad (27)$$

where w_i^* and r_i are the weights and abscissae of the selected quadrature rule of order α in the interval $[0, r^*]$. Usually, quadrature rules can be taken in the interval $[0, 1]$. In such a case, making the change of variable: $t = r/r^*$, eq. (27) can be rewritten as:

$${}_k\omega_l^m = r^{*l+3/2} \sum_{i=1}^{\alpha} w_i \mathcal{R}_{kl}(t_i) t_i^{l+2} f_{lm}(r^* t_i) \quad (28)$$

where w_i and t_i are the weights and abscissae of the quadrature rule of order α in the interval $[0, 1]$, and the relation: $w_i^* = r^* w_i$ has been taken into account.

Expansion of density in Canterakis–Zernike functions

The candidate for the expansions considered in this work is the set of Canterakis–Zernike functions proposed by Canterakis.^[18] Following this author, these functions are defined by:

$$Z_{nl}^m(\mathbf{t}) = [\sqrt{3} \sum_{\nu=0}^k q_{kl}^{\nu} t^{2\nu+l}] \mathcal{N}_{lm}^{\Omega} z_l^m(\theta, \varphi) \quad (29)$$

$k = (n-l)/2$ being an integer (i.e. $n-l$, an even number), and the coefficients q_{kl}^{ν} are:

$$q_{kl}^{\nu} = \frac{(-1)^k}{2^{2k}} \sqrt{\frac{2l+4k+3}{3}} \binom{2k}{k} (-1)^{\nu} \frac{\binom{k}{\nu} \binom{2(k+l+\nu)+1}{2k}}{\binom{k+l+\nu}{k}} \quad (30)$$

It can be proved that these functions are orthonormal in the unit ball, i.e. they fulfill eq. (11). Therefore, identifying: $\mathcal{F}_{kl}^m(\mathbf{t}) \equiv Z_{nl}^m(\mathbf{t})$ in the equations for the general expansion, it follows immediately that:

$${}_k\tilde{\mathcal{Z}}_l^m(\mathbf{r}) \equiv \frac{1}{\sqrt{r^{*3}}} Z_{nl}^m(\mathbf{t}) \quad (31)$$

are orthonormal in the interval $[0, r^*]$, with radial functions:

$$\tilde{\mathcal{Z}}_{kl}(r) = \sqrt{\frac{3}{r^{*3}}} \sum_{\nu=0}^k q_{kl}^{\nu} (t)^{2\nu+l} \quad (32)$$

where the index n has been kept in the definition of Z functions to follow Canterakis original notation, but it has been replaced by k in the $\tilde{Z}$ functions to adapt the notation to that of the previous section. (Notice that no ambiguity would arise if index k is taken in the l.h.s. of eq. (29) instead of n . Only historical reasons justify to keep index n in the definition.)

With these definitions, the expansion of the electron density ρ in a ball of radius r^* is obtained by replacing ${}_k\tilde{\mathcal{F}}_l^m(\mathbf{r})$ by ${}_k\tilde{Z}_l^m(\mathbf{r})$ in eq. (14). In practice, the summations on k and l indices must be truncated to given orders, K and L , respectively, thus determining the number of moments to be computed, as shown in eq. (33).

$$\rho(\mathbf{r}) \simeq \sum_{k=0}^K \sum_{l=0}^L \sum_{m=-l}^l {}_k\omega_l^m {}_k\tilde{Z}_l^m(\mathbf{r}) \quad (33)$$

and, according to eq. (28), the expansion coefficients are given in this case by:

$${}_k\omega_l^m = \langle \tilde{Z}_l^m | \rho \rangle_{r^*} = r^{*l+3/2} \sum_{i=1}^{\alpha} w_i \left[\sqrt{3} \sum_{\nu=0}^k q_{kl}^{\nu}(t_i)^{2\nu+l} \right] t_i^{l+2} f_{lm}(r^* t_i) \quad (34)$$

These coefficients, ${}_k\omega_l^m$, are known as *Canterakis–Zernike moments* in the literature, and their calculation by means of eq. (34) requires the evaluation of the polynomial $S_{kl}(t) = \sqrt{3} \sum_{\nu=0}^k q_{kl}^{\nu} t^{2\nu+l}$ at quadrature points t_i . In this regard, it is important to notice that direct computation of $S_{kl}(t)$ may lead to numerical issues for large values of k and l . To clarify this point, we illustrate in Table 1 the computation of $S_{kl}(t)$ for the particular case of $k=l=15$ and $t_{\text{ref}} = \frac{1}{2}(1 + \cos(633\pi/4096)) \simeq 0.942219719359127$. The middle column collects the summands, and in the right-most we can follow along the progression of the partial sums, applying Horner's method, and accompany its convergence in comparison with a reference value of $S_{1515}(t_{\text{ref}}) = -0.01818338110010633$, computed with Mathematica^[33] using very high precision.

As the table clearly shows, the loss in numerical accuracy due to cancellations eventually may affect, as in this case, all meaningful figures in double precision calculations. To prevent these accuracy losses, the computation can be made by means of recursion formulas. For this purpose, the right-hand side of eq. (32) can be rewritten as:

$$\sqrt{\frac{3}{r^{*3}}} \sum_{\nu=0}^k q_{kl}^{\nu} t^{2\nu+l} = \frac{(-1)^k}{2^{2k}} \sqrt{\frac{2l+4k+3}{r^{*3}}} \binom{2k}{k} t^l g_{kl}(t) \quad (35)$$

with

$$g_{kl}(t) \equiv \sum_{\nu=0}^k (-1)^{\nu} \frac{\binom{k}{\nu} \binom{2(k+l+\nu)+1}{2k}}{\binom{k+l+\nu}{k}} t^{2\nu} \quad (36)$$

The functions g_{kl} fulfill the following recurrence relation:

$$g_{k+1,l}(t) = \frac{4k+2l+3}{(2k+2l+3)(2k+1)} \left[\frac{4k(2k+2l+3)+4l(l+2)+3}{4k+2l+1} - (4k+2l+5)t^2 \right] g_{k,l}(t) - \frac{4k^2(4k+2l+5)(2k+2l+1)}{(4k+2l+1)(2k+2l+3)(2k-1)(2k+1)} g_{k-1,l}(t) \quad (37)$$

which can be started with $g_{-1,l}(t) = 0$ and $g_{0,l}(t) = 1$.

Furthermore, to avoid usage of large factorial numbers in eq. (35), we define the auxiliary quantities:

$$a_k \equiv (-1)^k \frac{1}{2^{2k}} \binom{2k}{k} = \frac{(-1)^k (k-1/2)!}{\sqrt{\pi} k!} \quad (38)$$

which follow a simple recursion: $a_{k+1} = -(2k+1)a_k/(2k+2)$, starting with $a_0 = 1$.

In summary, for computational purposes, it is convenient to write Canterakis–Zernike functions in the ball of radius r^* by combining eqs. (29)–(38), as:

$${}_k\tilde{Z}_l^m(\mathbf{r}) = \sqrt{\frac{2l+4k+3}{r^{*3}}} a_k g_{kl}(t) \mathcal{N}_{lm}^{\Omega} z_l^m(\mathbf{t}) \quad (39)$$

where $z_l^m(\mathbf{t})$ are the regular spherical harmonics in the unit ball, defined by: $z_l^m(\mathbf{t}) = t^l z_l^m(\theta, \varphi)$, which could be easily computed also by recursion^[29] when required.

The application of this scheme in the case previously shown in Table 1 yields a value of $S_{1515}(t_{\text{ref}}) = -0.0181833811000777$ which compares very well with the reference value (twelve coincident significant figures).

It is worth mentioning that eq. (39) is also suitable for computation of derivatives of the expansion of density in Canterakis–Zernike functions with respect to spatial coordinates. Thus, for the first derivative with respect to coordinate x , it gives:

Table 1. Summands and partial sums in the evaluation of Canterakis–Zernike function with $k=l=15$ and $t_{\text{ref}} = (1 + \cos(633\pi/4096))/2$ using Horner's method

j	$\sqrt{3} q_{1515}^{\nu} t_{\text{ref}}^{2\nu+15}$	$\sqrt{3} \sum_{\nu=0}^j q_{1515}^{\nu} t_{\text{ref}}^{2\nu+15}$
0	−0.855947805471435E+09	−0.855947805471435E+09
1	0.217605330540804E+11	0.209045852486090E+11
2	−0.251140792550383E+12	−0.230236207301774E+12
3	0.174951155434205E+13	0.151927534704028E+13
4	−0.824379022194130E+13	−0.672451487490102E+13
5	0.278822927673354E+14	0.211577778924344E+14
6	−0.700383676354879E+14	−0.488805897430535E+14
7	0.133239689809552E+15	0.843591000664987E+14
8	−0.193789775032666E+15	−0.109430674966167E+15
9	0.215735581049451E+15	0.106304906083284E+15
10	−0.182512347160339E+15	−0.762074410770547E+14
12	0.115338997047751E+15	0.391315559706965E+14
12	−0.527491578537596E+14	−0.136176018830631E+14
13	0.164946159507952E+14	0.287701406773217E+13
14	−0.31556338145157E+13	−0.278619313719399E+12
15	0.278619313719465E+12	0.66E−01
Exact	−	−1.818338110010633E−02

$$\begin{aligned}\frac{\partial_k \tilde{Z}_l^m(\mathbf{r})}{\partial x} &= \sqrt{\frac{2l+4k+3}{r^{*3}}} a_k \mathcal{N}_{lm}^{\Omega} \left[\frac{\partial g_{kl}(t)}{\partial x} z_l^m(\mathbf{t}) + g_{kl}(t) \frac{\partial z_l^m(\mathbf{t})}{\partial x} \right] \\ &= \sqrt{\frac{2l+4k+3}{r^{*3}}} a_k \mathcal{N}_{lm}^{\Omega} \frac{1}{r^*} \left[x \frac{dg_{kl}(t)}{dt} z_l^m(\mathbf{t}) + g_{kl}(t) \frac{\partial z_l^m(\mathbf{t})}{\partial t_x} \right]\end{aligned}\quad (40)$$

where it has been taken into account that

$$\frac{\partial t}{\partial x} = \frac{\partial(r/r^*)}{\partial x} = \frac{1}{r^*} \frac{x}{r'}$$

and the chain rule has been used:

$$\frac{\partial}{\partial x} = \frac{dt_x}{dx} \frac{\partial}{\partial t_x} = \frac{d}{dx} \left(\frac{x}{r^*} \right) \frac{\partial}{\partial t_x} = \frac{1}{r^*} \frac{\partial}{\partial t_x}$$

The derivatives of the regular spherical harmonics with respect to Cartesian coordinates can be computed in a straightforward way in terms of regular spherical harmonics.^[34]

Similar expressions are attained for the derivatives with respect to coordinates y and z .

Finally, the derivatives of the $g_{kl}(t)$ functions can be computed by recursion. Thus, taking the derivative of eq. (37) with respect to t yields:

$$\begin{aligned}g'_{k+1,l}(t) &= \frac{4k+2l+3}{(2k+2l+3)(2k+1)} \\ &\left[\frac{4k(2k+2l+3)+4l(l+2)+3}{4k+2l+1} - (4k+2l+5)t^2 \right] g'_{k,l}(t) \\ &- \frac{4k^2(4k+2l+5)(2k+2l+1)}{(4k+2l+1)(2k+2l+3)(2k-1)(2k+1)} g'_{k-1,l}(t) \\ &- \frac{(4k+2l+3)(4k+2l+5)}{(2k+2l+3)(2k+1)} 2t g_{k,l}(t)\end{aligned}\quad (41)$$

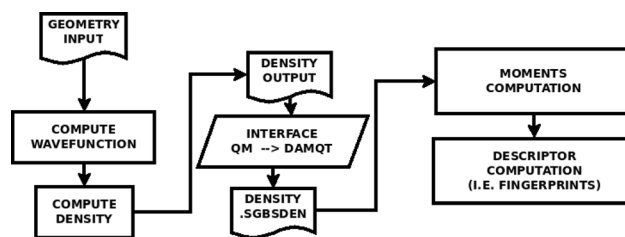


Figure 3. Flowchart illustrating the steps involved in obtaining the Canterakis–Zernike moments from a molecular geometry using any supported Quantum Chemistry software.

which starts with $g'_{-1,l}(t)=0$, $g'_{0,l}(t)=0$ and $g_{0,l}(t)=1$. The short notation $g'_{k,l}(t) \equiv (dg_{k,l}(t))/dt$ has been used for brevity.

Results and Discussion

The decomposition of the LCAO electron densities shown here was coded in Fortran 90/95 and has been implemented in the electron density analysis software DAMQT^[34,35] for testing. Its GUI is shown in Figure 2.

Besides the preliminar implementation in DAMQT, to facilitate visualization with its built-in 3D viewer, a standalone version of the program has been prepared, which is more appropriate to automate the decomposition of a large number of densities. The code was compiled with GNU Fortran Compiler and MPI was used for parallelization.

The standalone package also includes the interfaces to convert between relevant output files from some quantum chemistry software packages into the unified input file formats used by our software to compute the decompositions. A summary of the steps involved is shown in Figure 3.

The final implementation of the software is composed of three steps that are carried out sequentially:

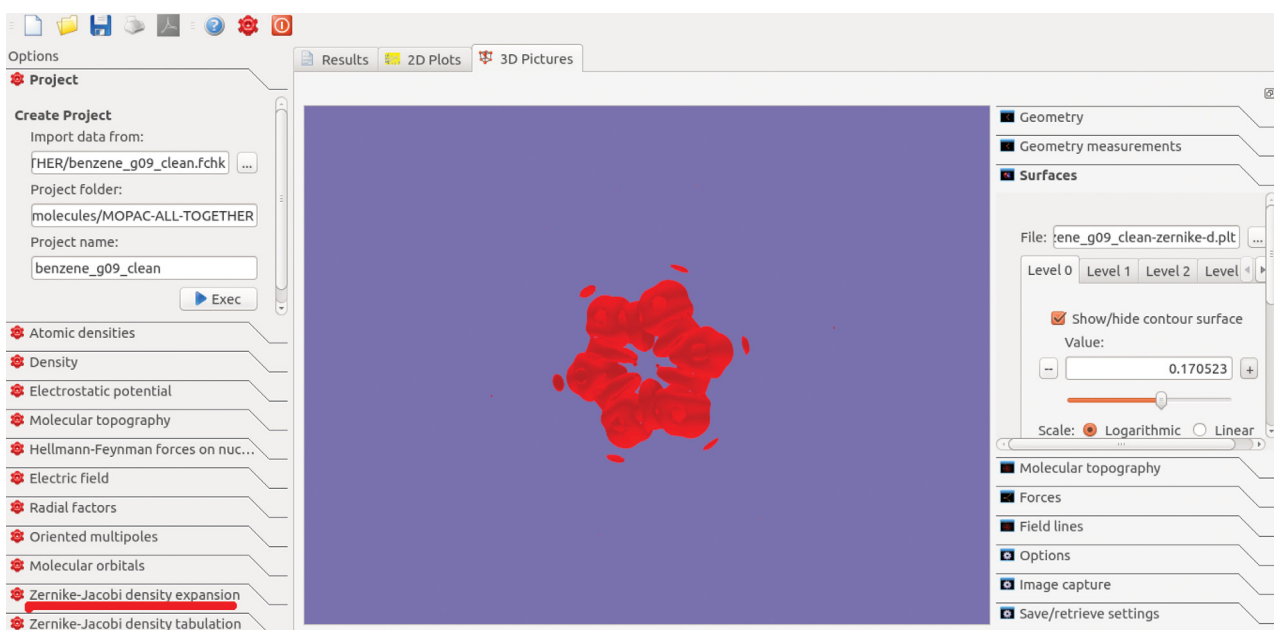


Figure 2. Screenshot of Canterakis–Zernike expansion of benzene in DAMQT Graphical User Interface. [Color figure can be viewed at wileyonlinelibrary.com]

Table 2. Computation time (in sec) for the three steps of the algorithm ($L = 20$, $K = 40$ and 512 quadrature points in all cases)

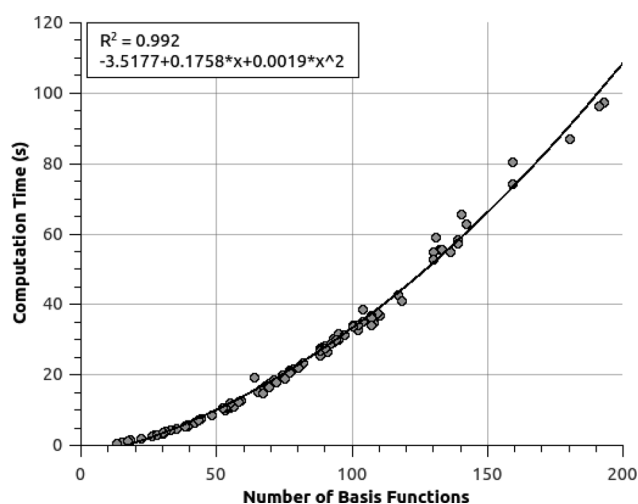
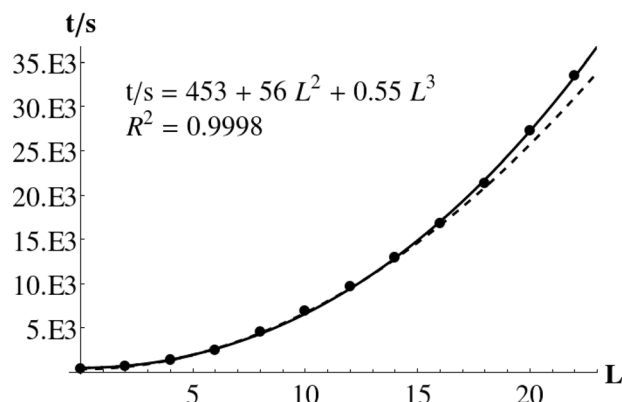
System	Basis	Size[a]	Step 1	Step 2	Step 3
Acetylene	6-311g	32	2.9	0.016	0.001
Acetic acid	6-311g	64	9.5	0.019	0.001
Cubane	6-311g	128	35.6	0.018	0.001
Acetylene...F ₂	Aug-CC-pVTZ	230	494.4	0.017	0.001
TWD1[b]	TZVP	6582	27315	0.021	0.001

[a] Number of basis functions. [b] System 1-N-naphthylphthalamic acid-Arabadopsis ABCB chaperone, TWISTED DWARF1 (TWD1).^[36]

1. Computation, at the quadrature points, of the radial factors that go into eq. (23) from those in eq. (19) with the algorithm previously described.
2. Computation of the Canterakis–Zernike Moments by numerical integration (quadrature) as shown in eq. (34).
3. Computation of the Rotationally Invariant Fingerprints according to eq. (18).

We have analyzed the performance of the algorithm in terms of the computational cost of each step and its relationship with the size of the problem.

With respect to the time required in each step, Table 2 clearly shows that the first step is by far the most time consuming, the cost of steps 2 and 3 being negligible even in the less favorable case of small systems, as these steps do not depend on the number of basis functions. In case of really big systems, time of steps 2 and 3 is completely negligible, as shown in the last entry of the table, which corresponds to the pocket-ligand system 1-N-naphthylphthalamic acid-Arabadopsis ABCB chaperone, TWISTED DWARF1 (TWD1),^[36] with 525 atoms and geometry optimized with a TZVP basis set^[37] by Prof. Noel Ferro, who kindly supplied us the data. Timing in table was measured on a system with an Intel(R) Xeon(R) CPU E5-2630 v4 @ 2.20GHz processor.

**Figure 4.** Scaling of computational time with the number of basis functions ($L = 10$, 512 quadrature points).**Figure 5.** Scaling of computational time with the length of expansion in spherical harmonics given by index L . ($K = 40$ and 512 quadrature points in all cases). Solid line: cubic fit. Dashed line: quadratic limit.

According to these considerations, we have analyzed the dependence of computational cost of step 1 with the number of basis functions, the length of spherical harmonics expansion, determined by index L , and the number of quadrature points. This step does not depend on the index K which determines the number of Canterakis–Zernike functions used for expanding each radial factor.

Figure 4 clearly shows that the cost of step 1 varies as the square of the number of basis functions. This is also the dependence predicted according to the code structure. Nevertheless, for very large systems, in which a significant number of two-center charge distributions will be negligible, a weaker dependence (tending to linear) is expected to occur.

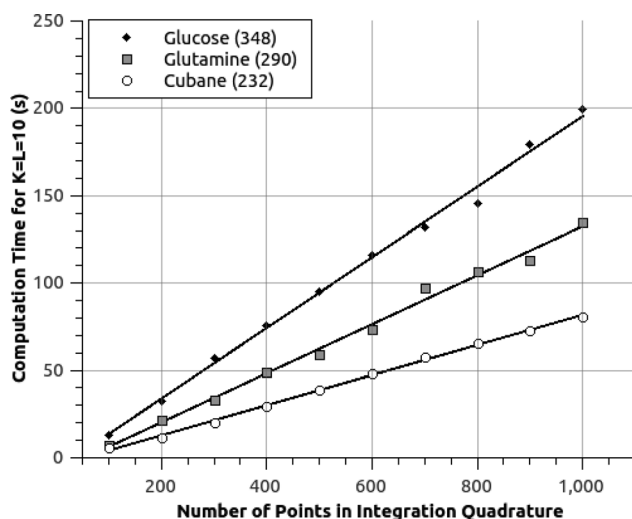
The dependence with the length of the spherical harmonics expansion, as it can be seen in Figure 5, is cubic on L , as predicted from the algorithm structure. However, for moderate values of L (up to 15) it is almost quadratic (see dashed line in figure), i.e. cubic terms, which are related to rotations, only become significant for large expansions.

Finally, the time in this step increases linearly with the number of quadrature points, as depicted in Figure 6. As mentioned before, time does not depend on K .

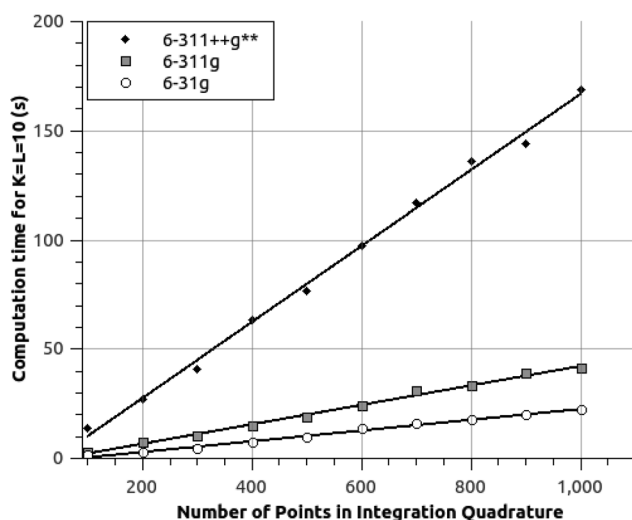
Steps 2 and 3 depend quadratically on L and in K , but they are so fast, as Table 2 illustrates, that their contribution to total computation time is completely negligible.

It is worth mentioning the fact that our procedure employs a one-dimension quadrature, unlike procedures requiring 3D numerical integrations. This facilitates the improvement of the accuracy in moments and fingerprints at a rather moderate cost.

Because of the relatively high cost of step 1 for big systems, we have carried out a parallelization with MPI. The performance of parallelization has been tested in the pocket-ligand system TWD1 above mentioned. Expansions with $L = 10$ and $K = 40$ were obtained in all cases with 1024 quadrature points. In Figure 7, wall clock time for step 1 is plotted against the reciprocal of the number of processors, as well as the corresponding line fit. It can be seen that computational cost reduction with the number of processors is virtually ideal: wall clock time performs as a fixed constant plus a term which decreases as the



(a)



(b)

Figure 6. Time scaling with the number of quadrature points a) for three different molecules (6-311++G**), b) for different number of basis sets (Glucose). $L = 10$ in all cases.

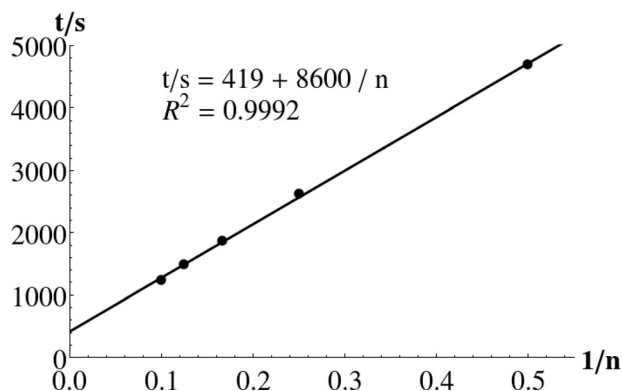


Figure 7. Wall clock time scaling with the reciprocal of the number of processors n .

reciprocal of the number of processors. Furthermore, time is quite evenly shared among processors, the difference between the highest and lowest times ranging from 0.6%, when two or four processors are used, to 6% for ten processors.

The results of the procedure were tested by comparing the multipole moments computed with the expansion in Canterakis–Zernike functions with the values obtained directly from the basis set and the density matrix, as well as analyzing the density reconstruction in several systems.

In Table 3, the multipole moments up to $L = 3$ computed from density matrix and basis set are compared with the values obtained from Canterakis–Zernike expansions obtained with Chebyshev quadratures of three different lengths. It can be seen that with 1024 quadrature points, even in the less accurate case which corresponds to dipole components, at least one correct figure is attained. This may seem too little, but considering the large number of contributions coming from the almost four millions of non-negligible¹ charge distributions (out of more than twenty one millions of total distributions), and the cancellations that must occur to yield the relatively small values of those components, the fact that not only the sign and the order but also some correct figures are attained can be considered as a good indicator of the correctness of the codes and the stability of the algorithm.

To give an idea of the performance of the method, we will mention that the computation of Canterakis–Zernike expansions up to $L = 10$ and $K = 40$ in the TWD1 system, with 1024 quadrature points and using 10 parallel threads, required a wall clock time of 1260 sec measured on a computer with 10 cores Intel(R) Xeon(R) CPU E5-2630 v4 @ 2.20GHz. The average computation time among processors was 1233 sec, with a standard deviation of 26 sec, and a difference between the highest and lowest times of 82 sec (6.6% of the average time). Considering the number of non-negligible charge distributions and the number of quadrature points, this time corresponds to an average cost for the computation of the contributions to the whole set of radial factors (up to $L = 10$) of 3 microseconds per non-negligible distribution and quadrature point (0.3 microseconds of wall clock time for 10 processors).

Finally, to illustrate how the expansion of density in Canterakis–Zernike functions approaches the actual density, Figure 8 shows a 6-311++G** electron density of a Crown Ether with four oxygen atoms and its reconstruction in terms of Canterakis–Zernike functions using an expansion with $K = L = 20$ and using a Chebyshev quadrature with 1024 points. As expected, the expansion captures the shape of the density at medium values, reproducing overall features rather than detailed nuances.

Conclusions

A new method has been reported for the computation of the Canterakis–Zernike Moments and related rotationally invariant fingerprints of theoretical electron densities coming from LCAO

¹ In this work, distributions are considered as negligible if the sigma overlap of the normalized basis functions (i.e., overlap excluding the normalized angular functions) is lower than 10^{-12} .

Table 3. Multipole moments^[a] of system TWD1 computed from Canterakis–Zernike expansions with 512, 1024 and 2048 quadrature points

<i>l</i>	<i>m</i>	Reference ^[b]	C–Z 512	C–Z 1024	C–Z 2048
0	0	–0.19800E+04	–0.19798E+04	–0.19800E+04	–0.19800E+04
1	–1	0.37657E+01	–0.08168E+01	0.36078E+01	0.35871E+01
	0	–0.30711E+01	0.05187E+01	–0.27859E+01	–0.31418E+01
	1	0.82157E+01	1.48312E+01	0.88683E+01	0.87830E+01
2	–2	0.15228E+05	0.15203E+05	0.15227E+05	0.15227E+05
	–1	0.51423E+04	0.51177E+04	0.51329E+04	0.51327E+04
	0	–0.13528E+06	–0.13533E+06	–0.13528E+06	–0.13529E+06
	1	–0.45035E+05	–0.44989E+05	–0.45031E+05	–0.45031E+05
	2	–0.62547E+04	–0.62498E+04	–0.62536E+04	–0.62547E+04
3	–3	0.29224E+05	0.29159E+05	0.29212E+05	0.29217E+05
	–2	0.59621E+05	0.59378E+05	0.59576E+05	0.59581E+05
	–1	–0.47886E+06	–0.47901E+06	–0.47897E+06	–0.47898E+06
	0	–0.49192E+06	–0.49275E+06	–0.49181E+06	–0.49180E+06
	1	0.30078E+06	0.30115E+06	0.30081E+06	0.30082E+06
	2	0.20662E+05	0.20801E+05	0.20714E+05	0.20707E+05
	3	0.22828E+05	0.22822E+05	0.22826E+05	0.22826E+05

[a] Defined as the factors that multiply the unnormalized real spherical harmonics in the expansion of the electron contribution to the long-range potential, except in case of $l = 0$ which refers to electron charge. [b] Computed from basis set, geometry and density matrix.

calculations at any level of computation. The method differs from other existing procedures because it assumes that the scalar field to be decomposed has the structure of an LCAO electron density, and fully exploits this fact, in combination with a previously reported method for translation of charge distributions, to yield a very efficient algorithm which requires a one-dimension numerical integration in a finite interval, which can be carried out with any suitable quadrature rule.

The computational cost of the code scales quadratically with the number of basis functions, and with the third power of the length of the expansion in spherical harmonics determined by index L . However, for values of L up to 15, this scaling is practically quadratic. It also scales linearly with the number of quadrature points, and is independent on the index K which determines the number of Canterakis–Zernike functions used for expanding each radial factor.

The stability and accuracy of the method has been tested by comparing the values of the multipole moments obtained with the Canterakis–Zernike expansions of different lengths with those directly computed from density matrix, basis set and geometry. This analysis has been complemented with the reconstruction of the electron density from the expansion, which has been visualized and compared with the original density by means of the 3D viewer implemented in DAMQT package.

Furthermore, the method itself has been implemented in the DAMQT software and can be applied as it is to the output files of the main quantum chemistry suites currently available. An MPI version of the code has also been developed for use with large systems. The program is actively maintained and up to date with current advances, with an active community of supporters. Our current perspectives already include plans to port currently used tools for Canterakis–Zernike expansions into the software for large scale usage as well as the implementation of new methods of our own.

With this work, we have achieved a computational framework that is accurate and fast, and which allows users of commonly employed quantum chemistry software to make use of all the applications that have been developed around the Canterakis–Zernike functions, presented in the “Introduction” section.

The existence of a link between the usually employed quantum chemistry software systems and the applications employing the Canterakis–Zernike moments makes it easier to collect and make available to the community all the methods and algorithms which employ these moments and their derived quantities.

A copy of the most recent version of DAMQT package, which includes the programs for Canterakis–Zernike expansions of density reported herein can be freely downloaded from Mendeley repository, at <https://data.mendeley.com/datasets/rpzbrppywy/3> (DOI: 10.17632/rpzbrppywy.3).

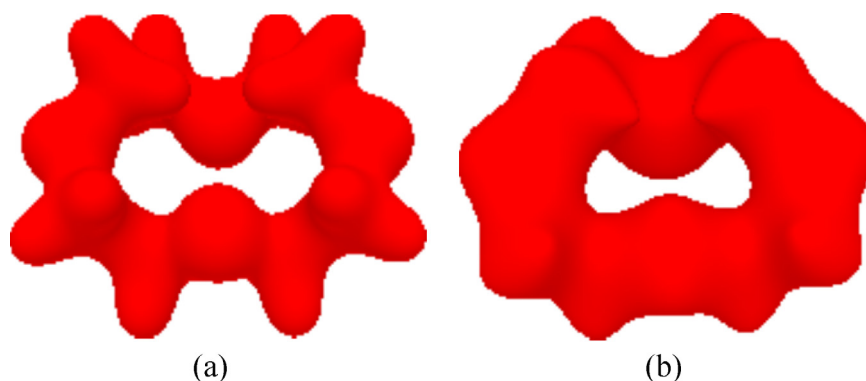


Figure 8. a) Crown Ether original Gaussian 6-311++G** density and b) density reconstructed by an expansion of Canterakis–Zernike Functions generated from 1024 quadrature points, for $K=L=20$ at isolevel 0.1. [Color figure can be viewed at wileyonlinelibrary.com]

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Keywords: quantum molecular similarity · molecular fingerprints · molecular pattern recognition · Canterakis–Zernike moments · electron density expansion

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